

Flappy Bird - Project Overview

Project Flow

Player presses Space ↓ BirdScript applies upward velocity ↓ Bird passes between pipes ↓ pipeMiddleScripte adds score ↓ LogicScripte updates UI

If Bird hits pipe/ground OR leaves screen bounds ↓ BirdScript triggers Game Over ↓ LogicScripte displays Game Over Screen

Script Responsibilities

BirdScript

Purpose: Controls the player character.

Responsibilities:

- Detect Space key input
- Apply flap force using Rigidbody2D
- Detect collisions
- Detect leaving screen bounds
- Trigger Game Over
- Disable actions after death

Main Variables:

- flapStrenghtth
 - BirdIsAlive
 - myRigidbody
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LogicScripte

Purpose: Acts as the Game Manager.

Responsibilities:

- Store player score
- Update score UI
- Display Game Over screen
- Restart current scene
- Quit application

- Track game state using gameIsOver

Main Variables:

- PlayerScore
 - gameIsOver
 - gameOverScreen
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PipeSpawnerScript

Purpose: Creates pipe obstacles.

Responsibilities:

- Spawn first pipe
- Spawn pipes repeatedly
- Generate random pipe heights

Main Variables:

- SpawnRate
 - timer
 - heightOffset
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PipeMoveScripte

Purpose: Moves obstacles.

Responsibilities:

- Move pipes left
- Destroy pipes after leaving screen

Main Variables:

- PipeMoveSpeed
 - deadzone
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pipeMiddleScripte

Purpose: Handles scoring.

Responsibilities:

- Detect bird passing through pipe gap
- Increase score
- Prevent scoring after Game Over

Dependencies:

- LogicScripte
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Important Relationships

BirdScript — calls → LogicScripte.gameOver()

pipeMiddleScripte — calls → LogicScripte.addScore()

LogicScripte — controls UI

PipeSpawnerScript — creates Pipe Prefabs

PipeMoveScripte — destroys old Pipe Prefabs

Known Improvements

Future Version Ideas:

- High Score system
 - Sound effects
 - TextMeshPro UI
 - Pause menu
 - Difficulty scaling
 - Background themes
 - Object Pooling for pipes
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Bugs Already Fixed

- Score increasing after Game Over
- Game Over screen visible on startup
- UI scaling issues after Unity upgrade
- Bird gravity tuning